

Vignesh Kumar P

Software Engineer

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SUMMARY

Engineering student specializing in Python development, machine learning, and edge computing, with hands-on experience building real-time embedded systems, reinforcement learning agents, and CNN-based defect detection models. Published researcher with proven ability to design, train, and deploy AI systems end-to-end. Skilled in Python, OpenCV, NumPy, LabVIEW, and explainable AI tools. Seeking a Python Developer or ML Engineer role where I can contribute to impactful, data-driven software solutions.

EXPERIENCE

Frontend Developer

Apr '25 — May '25

Infoem Software Solutions

Namakkal, India

- Completed a hands-on internship focused on Front-End Web Development using **HTML, CSS, and JavaScript**
- Designed and developed a responsive web project, applying core front-end concepts and best practices
- Gained practical exposure to real-world software development workflows in a professional environment

SKILLS

Languages & Libraries: Html, CSS, Python, MySQL, Javascript

ML / AI Frameworks NumPy, Matplotlib, OpenCV, MobilenetV3, Q-Learning, Gradcam, SHAP

Tools & Platforms VS Code, Raspberry Pi, Edge Computing, Github, Git, LabVIEW

PROJECTS

IoT-Based Fire Detection and Suppression System for Railway Safety

'25

- Aim:** Designed and implemented an automated fire detection and suppression system using LabVIEW for real-time monitoring and control, integrated with Arduino, temperature and smoke sensors, GSM module, and water pump to enhance railway safety.
- Tools:** Arduino UNO, LM35, MQ2, SIM800L, LabVIEW, IoT, Embedded C

Reinforcement Learning for Path Planning in Grid Environments using Q-Learning

'25

- Aim:** Developed a reinforcement learning agent using Q-Learning to perform autonomous navigation in 2D static and dynamic grid environments by learning optimal policies to avoid obstacles and reach target goals.
- Tools:** Python, Q-Learning, NumPy, Matplotlib, CSV, Grid World Simulation

Design of Edge Computing System for Steel Surface Defect Detection

'25

- Aim:** To design and deploy a lightweight CNN-based edge computing system for real-time steel surface defect detection with explainable AI to improve transparency and industrial trust.
- Tools:** Python, Lightweight CNN (MobileNetV3), Edge Computing (Raspberry Pi), Explainable AI (Grad-CAM, SHAP), OpenCV, Web Interface

EDUCATION

B.tech in Automation and Robotics Engineering, Amrita School of Engineering

Aug '22 — Apr '26
Coimbatore

PUBLICATIONS

IoT-Based Fire Detection and Suppression System for Railway Safety

Dec '25

ReACS

Designed and implemented an automated fire detection and suppression system using LabVIEW for real-time monitoring and control, integrated with Arduino, temperature and smoke sensors, GSM module, and water pump to enhance railway safety.

CERTIFICATIONS

Email Writing Skills, Infosys Spring Board

Jan '25

Front End Development, Infoem Software Solutions

May '25

AWS AI Practitioner Challenge, Udacity

May '26

Data Analytics Job Simulation, Deloitte

May '26